

Five-Token Dynamics Beyond Lane-Order Symmetry

Local Dihedral Closure and a Complete Single-Lane Permutation Dichotomy

WuJun Chen

Independent Researcher

2026

This paper (Paper XXXII of the RIME program) studies intrinsic raw spectator-safe return dynamics beyond lane-order symmetry. It imports the normalized partial-return system and exact path simulation of Paper XXXI, but no typed transfer, projectability, recursive-return, credit-settlement, or reset-bound conclusion.

Abstract

Problem. Paper XXXI reduces a binary-kernel single-defect return system to a partial system on two marked tokens, three spectators, and the remaining holes, with an arbitrary branch permutation $a \in \text{Sym}(E)$. For normalizer branches, lane adjacency is sufficient for Safe-Hit and same-lane incidence is necessary. The unresolved region consists of same-lane configurations whose marked tokens are not adjacent after holes are deleted.

Approach. We first isolate the exact geometric condition used by the negative direction: the branch permutation must carry every arithmetic lane to a lane by a cycle-graph isomorphism. We then restrict to the one-lane regime $\gcd(n, \Delta) = 1$. A state with fixed hole-deleted token order is encoded by a weak composition

$$(h_0, \dots, h_4), \quad h_i \geq 0, \quad \sum_i h_i = n - 6.$$

A guarded identity-branch return transfers one hole from a gap to the preceding gap. These directed transfers are strongly connected on every fixed weak-composition fiber.

Results. Lane-wise cycle-graph isomorphisms introduce no new Safe-Hit states. This includes affine normalizer branches with mixed Chinese-remainder signs not covered by one global ± 1 congruence. In the one-lane regime, identity-branch dynamics is transitive on every hole-deleted five-token order fiber. Let C_L be the physical lane cycle, with $L = n - 1$. For every branch permutation $a \in \text{Sym}(E)$,

$$\text{Hit}(a) = \begin{cases} \text{Adj}, & a \in \text{Aut}(C_L), \\ \mathcal{Z}_\Delta, & a \notin \text{Aut}(C_L). \end{cases}$$

Thus every non-dihedral single-lane branch permutation makes all marked five-token configurations spectator-safely target-reachable.

Boundary. The theorem does not classify the multi-lane regime. For a branch preserving the arithmetic lane partition, order breaking may occur on only one lane, and a proof would need a lane-type-sensitive mobility theorem. The paper also does not classify survivor placement or lift raw Safe-Hit to typed transfer semantics.

Keywords. synchronizing automata; single-defect automata; partial dynamics; branch permutations; cycle automorphisms; Safe-Hit; guarded reachability

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/n\mathbb{Z}$	labelled circular carrier, $n \geq 6$
$p(q) = q + 1$	fixed full cycle
Δ	oriented kernel separation
$g = \gcd(n, \Delta)$	number of arithmetic lanes before puncturing
$\ell = n/g$	full lane length
$E = Q \setminus \{0\}$	normalized branch carrier
$\widehat{\Sigma} = \text{Conf}_{2,3}(E)$	two marked tokens and three spectators
ψ_Δ	punctured rotation on E
$H_\Delta = \langle \psi_\Delta \rangle$	globally enabled cyclic return subgroup
$\mathcal{Z}_\Delta = \widehat{\Sigma}/H_\Delta$	punctured-rotation orbit space
$R_r^{(0)}$	identity-branch relation-valued quotient edge
$W_\Delta = N_{\text{Sym}(E)}(H_\Delta)/H_\Delta$	affine twist quotient
$\mathcal{C}_\Delta$	arithmetic-lane partition of E
$\text{Stab}(\mathcal{C}_\Delta)$	permutations carrying each lane to a lane
Adj_Δ	lane-adjacent orbit set
Same_Δ	same-lane orbit set
$\text{Hit}(a)$	orbit-wise Safe-Hit truth set for branch a
$\Omega(x)$	hole-deleted cyclic P/S word
$L = n - 1$	unique lane length when $g = 1$
C_L	physical cycle graph induced by ψ_Δ in the one-lane regime

1 Introduction

Paper XXXI normalizes every labelled partial return of a binary-kernel rank- $(n - 1)$ defect into the point map

$$\phi_r = \varepsilon_\Delta p^r a \quad (1.1)$$

on a carrier with two marked tokens, three spectators, and holes [4]. The formula transports the collision guard, the normalized exit target, and labelled paths exactly. When the branch permutation a normalizes the punctured-rotation group

$$H_\Delta = \langle \psi_\Delta \rangle, \quad (1.2)$$

the quotient dynamics is a finite relation-valued skew system. Paper XXXI proves the general sandwich

$$\text{LaneAdj}_\Delta(x) \implies \text{SafeHit}_{n,\Delta,a}(x) \implies \text{SameLane}_\Delta(x). \quad (1.3)$$

It also closes multipliers satisfying one global congruence $u \equiv \pm 1$ modulo the order of ψ_Δ .

Two questions remain hidden inside that formulation.

First, normalizer membership is stronger than the geometric condition used by the negative proof. It is enough that a branch permutation carry every arithmetic lane to a lane by a cycle-graph isomorphism. For affine normalizer branches this includes mixed Chinese-remainder signs that preserve order on one lane length and reverse it on the other.

Second, a branch permutation that is not an automorphism of the physical lane cycle sends some nonedge to an edge. The smallest regime in which the guarded consequence can be isolated is the one-lane case

$$g = \gcd(n, \Delta) = 1. \quad (1.4)$$

There every state is same-lane. The entire open part of (1.3) is therefore the nonadjacent token-order class.

The paper proves that this class is completely promoted by every non-dihedral branch permutation. The dynamical ingredient is a source-preserving mobility theorem. For a fixed cyclic order of the five occupied tokens, record the numbers of holes in the five intervening gaps. A guarded return moves one hole one step around the gap cycle. Since the reverse unit move is a four-step forward circuit, the resulting directed weak-composition graph is strongly connected. This turns the elementary nonedge-to-edge witness into an actual guarded path.

The theorem spine is:

1. Section 2 freezes the exact branch system inherited from Paper XXXI.
2. Section 3 closes every branch that is dihedral on each arithmetic lane.
3. Section 4 proves single-lane order-fiber transitivity by guarded hole transfers.
4. Section 5 combines that mobility with cycle-edge conversion to classify every single-lane branch permutation.
5. Section 6 records the lane-stabilizer sandwich and the precise multi-lane boundary.

The work remains intrinsic and raw. It does not reconstruct a defect, classify survivor placement, or import downstream typed authority.

1.1 Related Work and Novelty Boundary

The Černý conjecture and the broader theory of synchronizing automata concern short reset words [3, 13, 14]. Circular automata and almost-group automata provide nearby structural settings [1, 2, 8]. This paper proves no reset bound and no classification of general almost-group automata.

Monotonic automata, subset synchronization, careful synchronization, and dynamic word constraints study related order or partiality phenomena [9, 12, 15]. Complete reachability asks which unlabelled subsets can occur as images [7, 10, 11]. The present condition is different: one guarded path must preserve injectivity of the marked pair and all three spectators at every step and must hit a source-marked target.

Within the RIME line, Paper XXIX proves the canonical raw rank-five frontier classification, Paper XXX identifies the universal return group and free branch parameter, and Paper XXXI derives the normalized partial system and the multi-lane cyclic-branch classification [4–6]. The present paper begins at the remaining branch-permutation frontier. Its new claims are lane-wise dihedral closure and a complete one-lane permutation Safe-Hit theorem.

2 The Inherited Branch System

Fix $n \geq 6$ and $1 \leq \Delta \leq n - 1$. Put

$$g = \gcd(n, \Delta), \quad \ell = \frac{n}{g}, \quad E = \mathbb{Z}/n\mathbb{Z} \setminus \{0\}. \quad (2.1)$$

Let

$$\widehat{\Sigma} = \text{Conf}_{2,3}(E) \quad (2.2)$$

be the states $x = (A, R)$ with $|A| = 2$, $|R| = 3$, and $A \cap R = \emptyset$. The points of A carry color P, the points of R carry color S, and all remaining points of E are holes.

Paper XXXI defines

$$\psi_\Delta = \varepsilon_\Delta p^\Delta|_E, \quad H_\Delta = \langle \psi_\Delta \rangle, \quad \mathcal{Z}_\Delta = \widehat{\Sigma}/H_\Delta. \quad (2.3)$$

The cycles of ψ_Δ are one punctured lane of length $\ell - 1$ and, when $g > 1$, $g - 1$ ordinary lanes of length ℓ .

Lane adjacency and same-lane incidence are constant on H_Δ -orbits and therefore define

$$\text{Adj}_\Delta \subseteq \text{Same}_\Delta \subseteq \mathcal{Z}_\Delta. \quad (2.4)$$

2.1 Arbitrary branch permutations

For every $a \in \text{Sym}(E)$, the inherited normalized partial returns have point maps

$$\phi_r^{(a)} = \varepsilon_\Delta p^r a, \quad \widehat{\mathcal{T}}_{\Delta,a} = a_\#^{-1} \widehat{\mathcal{T}}_{\Delta,\text{id}}, \quad (2.5)$$

with the collision guard transported exactly as in Paper XXXI. The branch need not normalize H_Δ .

We use the following consequence of Paper XXXI repeatedly.

Lemma 2.1 (pathwise identity simulation).

Let $a \in \text{Sym}(E)$. Every finite guarded identity-branch path

$$x \xrightarrow{r_1, \dots, r_k}_{\text{id}} y \quad (2.6)$$

has a finite guarded realization from x to the same state y in the a -system.

Proof. Let $m = \text{ord}(a)$. The label-zero return in the a -system is the globally enabled permutation $a_\#$. Before simulating an identity edge with label r , apply the block 0^{m-1} and then r . At the point-map level this is

$$(\varepsilon_\Delta p^r a)_\# (a^{-1})_\# = (\varepsilon_\Delta p^r)_\#. \quad (2.7)$$

Its guard is exactly the guard of the identity edge being simulated. Concatenating these blocks realizes the complete path. This is the pathwise content of the identity-branch simulation theorem of Paper XXXI.

If $y \in \widehat{\mathcal{T}}_{\Delta,\text{id}}$, append one more block 0^{m-1} . It reaches $a_\#^{-1}y \in \widehat{\mathcal{T}}_{\Delta,a}$, so

$$\boxed{\text{SafeHit}_{n,\Delta,\text{id}}(x) \implies \text{SafeHit}_{n,\Delta,a}(x)}. \quad (2.8)$$

Powers of ψ_Δ give mutual identity-branch paths between all representatives of one H_Δ -orbit. Applying the lemma in both directions shows that Safe-Hit truth for arbitrary a is constant on each such orbit. Thus the primary truth sets of this paper are

$$\boxed{\text{Hit}(a) = \{\pi_H(x) : \text{SafeHit}_{n,\Delta,a}(x)\}, \quad \text{Prom}(a) = \text{Hit}(a) \cap (\text{Same}_\Delta \setminus \text{Adj}_\Delta)}. \quad (2.9)$$

Definition (2.9) does not assert that a induces a deterministic map on $\mathcal{Z}_\Delta$. It records orbit-wise truth only.

2.2 Normalizer specialization

For a normalizer branch

$$a \in N_{\text{Sym}(E)}(H_\Delta), \quad (2.10)$$

write $\bar{a}$ for its induced permutation of $\mathcal{Z}_\Delta$. If $R_r^{(0)}$ is the relation-valued identity-branch quotient edge, the exact skew edge is

$$\boxed{O \xrightarrow{r}_a O' \iff \bar{a}(O) R_r^{(0)} O'}. \quad (2.11)$$

The target-orbit set satisfies

$$\mathcal{T}_a = \bar{a}^{-1} \mathcal{T}_0, \quad (2.12)$$

and quotient reachability lifts exactly:

$$\text{SafeHit}_{n,\Delta,a}(x) \iff \pi_H(x) \rightsquigarrow_a \mathcal{T}_a. \quad (2.13)$$

No deterministic quotient map is asserted in (2.11). Put

$$W_\Delta = N_{\text{Sym}(E)}(H_\Delta)/H_\Delta. \quad (2.14)$$

Paper XXXI's cyclic-branch exponent-elimination theorem shows that quotient Safe-Hit depends only on $w \in W_\Delta$. For any representative a of w , define

$$\text{Hit}(w) = \{O \in \mathcal{Z}_\Delta : O \rightsquigarrow_w \mathcal{T}_w\} = \text{Hit}(a). \quad (2.15)$$

The inherited normalizer sandwich is

$$\boxed{\text{Adj}_\Delta \subseteq \text{Hit}(w) \subseteq \text{Same}_\Delta.} \quad (2.16)$$

3 Lane-Wise Dihedral Closure

In the affine lane coordinates of Paper XXXI, a normalizer representative has the form

$$\begin{aligned} a(*, t) &= (*, ut + \beta_*) \pmod{\ell - 1}, \\ a(j, t) &= (\sigma(j), ut + \beta_j) \pmod{\ell}. \end{aligned} \quad (3.1)$$

The global modulus used to encode the multiplier is

$$M = \begin{cases} \ell - 1, & g = 1, \\ \ell(\ell - 1), & g > 1. \end{cases} \quad (3.2)$$

The condition $u \equiv \pm 1 \pmod{M}$ is sufficient for preserving or reversing all lane orders, but it is not necessary when two lane lengths occur.

Let $\mathcal{C}_\Delta$ be the set of ψ_Δ -cycles and define its setwise stabilizer

$$\text{Stab}(\mathcal{C}_\Delta) = \{a \in \text{Sym}(E) : a(C) \in \mathcal{C}_\Delta \text{ for every } C \in \mathcal{C}_\Delta\}. \quad (3.3)$$

Each lane carries the cycle graph induced by its ψ_Δ order. Put

$$\text{LocalDih}_\Delta(a) \iff a \in \text{Stab}(\mathcal{C}_\Delta) \text{ and every } a|_C : C \longrightarrow a(C) \text{ is a cycle-graph isomorphism.} \quad (3.4)$$

For a normalizer representative in (3.1), condition (3.4) says that $t \mapsto ut$ preserves or reverses cyclic order modulo every lane length that occurs. For lengths at least three this is $u \equiv \pm 1$ at that length; for lengths at most two it is automatic. The signs on the punctured and ordinary lane lengths may differ.

Proposition 3.1 (lane-wise dihedral classification).

Let $a \in \text{Sym}(E)$. If $\text{LocalDih}_\Delta(a)$, then for every $x \in \widehat{\Sigma}$,

$$\boxed{\text{SafeHit}_{n,\Delta,a}(x) \iff \text{LaneAdj}_\Delta(x) = 1.} \quad (3.5)$$

Proof. The branch permutation a maps lanes to lanes and preserves or reverses their cyclic orders, so it preserves lane adjacency. Every guarded identity-branch return preserves

the same observable. Since an a -return is an identity-branch point map preceded by $a_{\#}$, every guarded a -return preserves lane adjacency. The target $\widehat{\mathcal{T}}_{\Delta,a} = a_{\#}^{-1}\widehat{\mathcal{T}}_{\Delta,\text{id}}$ is lane-adjacent, so adjacency is necessary. Conversely, the identity-branch classification of Paper XXXI makes every lane-adjacent state identity-hittable, and (2.8) simulates that witness in the a -system.

Proposition 3.1 is strictly broader than the affine normalizer statement. In that subfamily it removes a false dichotomy: a multiplier may fail the single global congruence $u \equiv \pm 1 \pmod{M}$ while remaining dihedral on each lane length by mixed Chinese-remainder signs. Such a branch creates no new Safe-Hit state.

4 Single-Lane Order-Fiber Transitivity

Assume for the rest of Sections 4 and 5 that

$$g = 1, \quad L = n - 1. \quad (4.1)$$

There is one ψ_{Δ} -lane, of length L . Since $n \geq 6$, we have $L \geq 5$. Every configuration is same-lane.

For $x \in \widehat{\Sigma}$, read the five occupied tokens cyclically along the ψ_{Δ} -lane and delete all holes. Denote the resulting cyclic P/S word by $\Omega(x)$. Temporarily distinguish the occupied positions in that cyclic order as

$$T_0, T_1, T_2, T_3, T_4. \quad (4.2)$$

Let h_i be the number of holes after T_i and before T_{i+1} , with indices modulo 5. Then

$$h_i \geq 0, \quad \sum_{i=0}^4 h_i = L - 5 = n - 6. \quad (4.3)$$

4.1 The local gap transfer

Lemma 4.1 (guarded unit gap transfer).

If $h_i > 0$, there is a guarded identity-branch path preserving $\Omega(x)$ and changing only

$$(h_{i-1}, h_i) \mapsto (h_{i-1} + 1, h_i - 1). \quad (4.4)$$

Proof. A power of the globally enabled return ψ_{Δ} places T_i at Δ . Since $h_i > 0$, the next lane position 2Δ is a hole. Apply the identity-branch return with label $r = -\Delta$.

The potential collision pair is occupied at Δ and empty at 2Δ , so the return is guarded. It fixes T_i , shifts the later positions of the following gap one Δ -step toward T_i , removes the first hole after T_i , and creates the compensating hole at the preceding lane position $-\Delta$. Explicitly, since $g = 1$, the points of E are $j\Delta$ for $1 \leq j \leq n - 1$, and

$$\phi_{-\Delta}(\Delta) = \Delta, \quad \phi_{-\Delta}(j\Delta) = (j - 1)\Delta \quad (2 \leq j \leq n - 1).$$

The second preimage of Δ , namely 2Δ , is the required hole, and $-\Delta$ is absent from the image support. Hence (4.4) holds. The lane-order preservation theorem of Paper XXXI shows that no occupied tokens cross. The token color is irrelevant to the guard and to the transfer.

Let

$$\mathcal{H}_N = \{(h_0, \dots, h_4) \in \mathbb{Z}_{\geq 0}^5 : h_0 + \dots + h_4 = N\} \quad (4.5)$$

with $N = L - 5$.

Lemma 4.2 (directed gap connectivity).

The directed moves

$$h \mapsto h + e_{i-1} - e_i \quad (h_i > 0) \quad (4.6)$$

make $\mathcal{H}_N$ strongly connected.

Proof. Each move transports one unit one step around the directed five-cycle of gap indices. Its apparent reverse, from gap $i - 1$ to gap i , is realized by transporting the same unit through the other four directed edges:

$$i - 1 \longrightarrow i - 2 \longrightarrow i - 3 \longrightarrow i - 4 \longrightarrow i \pmod{5}. \quad (4.7)$$

At every intermediate stage the gap currently carrying that unit is positive. Thus every elementary transfer between adjacent parts is reversible by a directed path. The ordinary graph of weak compositions with fixed total N is connected under adjacent unit transfers, so the directed graph is strongly connected. The case $N = 0$ is the single-vertex case.

Theorem 4.3 (single-lane order-fiber transitivity).

If $\Omega(x) = \Omega(y)$, then

$$x \xleftrightarrow{\text{id}} y \quad (4.8)$$

by guarded identity-branch returns.

Proof. Choose a cyclic identification of the five temporary token labels that respects the common color word. Lemmas 4.1 and 4.2 transform the source gap vector into the target gap vector without changing that cyclic token order. Once the gap vectors agree, a power of the regular cyclic return ψ_Δ aligns T_0 with its target coordinate. All other occupied coordinates and colors then agree automatically. Strong connectivity gives the reverse path.

The theorem is stronger than the hole-slide used in Paper XXXI. That argument decreased one selected marked-pair gap. Here every occupied token may serve as the transfer site, so the complete order fiber is connected.

5 The Single-Lane Permutation Dichotomy

Let C_L be the physical cycle graph on E induced by the regular ψ_Δ action. Its edge set is

$$\mathcal{E}_L = \{\{q, \psi_\Delta(q)\} : q \in E\}. \quad (5.1)$$

The group $\text{Aut}(C_L)$ is the set of branch permutations that preserve $\mathcal{E}_L$. We use this notation instead of D_L or D_{2L} to avoid dihedral-order conventions.

There are exactly two cyclic color-order classes:

$$[\text{PPSSS}] \quad \text{and} \quad [\text{PSPSS}]. \quad (5.2)$$

The first is lane-adjacent; the second is nonadjacent. By (2.9), $\text{Hit}(a)$ and $\text{Prom}(a)$ are defined on $\mathcal{Z}_\Delta$ for every $a \in \text{Sym}(E)$, even when a does not normalize H_Δ .

5.1 Cycle-edge conversion

For a permutation a of E ,

$$a \in \text{Aut}(C_L) \iff a(\mathcal{E}_L) = \mathcal{E}_L. \quad (5.3)$$

If $a \notin \text{Aut}(C_L)$, then $a(\mathcal{E}_L)$ and $\mathcal{E}_L$ are distinct sets of the same cardinality L . Hence some edge is missing from $a(\mathcal{E}_L)$. Taking its inverse image gives a two-set $B \subset E$ such that

$$\boxed{B \notin \mathcal{E}_L, \quad a(B) \in \mathcal{E}_L.} \quad (5.4)$$

Both open arcs of $C_L \setminus B$ are nonempty. Since $L \geq 5$, the three spectators can be placed with at least one on each arc. Thus there is a state y with

$$A_y = B, \quad \Omega(y) = [\text{PSPSS}], \quad (5.5)$$

while the globally enabled label-zero return $a_\#$ sends its marked pair to the physical edge $a(B)$. As before, the placement (5.5) is not itself a reachability proof; Theorem 4.3 supplies the actual guarded path to it.

Theorem 5.1 (single-lane permutation dichotomy).

Let $g = 1$ and $L = n - 1$. For every $a \in \text{Sym}(E)$,

$$\boxed{\text{Hit}(a) = \begin{cases} \text{Adj}_\Delta, & a \in \text{Aut}(C_L), \\ \mathcal{Z}_\Delta, & a \notin \text{Aut}(C_L). \end{cases}} \quad (5.6)$$

Proof. If $a \in \text{Aut}(C_L)$, there is one lane and a is lane-wise dihedral. Proposition 3.1 gives the first line.

Now suppose $a \notin \text{Aut}(C_L)$. Every adjacent orbit lies in $\text{Hit}(a)$ by (2.8) and the identity-branch classification. Let x be nonadjacent. Its token order is $[\text{PSPSS}]$. Choose B and y as in (5.4)–(5.5). Theorem 4.3 gives a guarded identity-branch path from x to y , and Lemma 2.1 simulates that path in the a -system, ending at the same state y .

Apply the globally enabled label-zero return. The resulting marked pair is the physical edge $a(B)$ and is therefore lane-adjacent. The identity-branch Safe-Hit theorem together with (2.8) supplies a guarded continuation to the a -target. Thus every nonadjacent orbit also lies in $\text{Hit}(a)$, proving the second line.

Corollary 5.2 (exact promotion set).

In the one-lane regime,

$$\text{Prom}(a) = \begin{cases} \emptyset, & a \in \text{Aut}(C_L), \\ \mathcal{Z}_\Delta \setminus \text{Adj}_\Delta, & a \notin \text{Aut}(C_L). \end{cases} \quad (5.7)$$

Proof. Subtract Adj_Δ from the two cases of Theorem 5.1 and use (2.9).

The theorem compares truth predicates. It does not assert equality of labelled path languages, equality of shortest witnesses, or a deterministic quotient of every partial return. In particular, the affine normalizer dichotomy is now only a specialization: an affine map $t \mapsto ut + \beta$ belongs to $\text{Aut}(C_L)$ exactly when $u \equiv \pm 1 \pmod{L}$.

6 Multi-Lane Boundary

When $g > 1$, the punctured lane has length $\ell - 1$ and ordinary lanes have length ℓ . The full lane-partition stabilizer has the natural form

$$\text{Stab}(\mathcal{C}_\Delta) \cong S_{\ell-1} \times (S_\ell^{g-1} \rtimes S_{g-1}). \quad (6.1)$$

The first factor acts on the unique punctured lane; the wreath-product factor acts independently inside the ordinary lanes and permutes those lanes. The normalizer $N_{\text{Sym}(E)}(H_\Delta)$ is the thinner subfamily in which all lane restrictions share one multiplier.

For $a \in \text{Stab}(\mathcal{C}_\Delta)$, define

$$\text{Break}_\Delta(a) = \{C \in \mathcal{C}_\Delta : a|_C : C \rightarrow a(C) \text{ is not a cycle-graph isomorphism}\}. \quad (6.2)$$

Proposition 3.1 closes the case $\text{Break}_\Delta(a) = \emptyset$. The full stabilizer also satisfies the same intrinsic sandwich.

Corollary 6.1 (lane-stabilizer Safe-Hit sandwich).

For every $a \in \text{Stab}(\mathcal{C}_\Delta)$ and $x \in \widehat{\Sigma}$,

$$\boxed{\text{LaneAdj}_\Delta(x) = 1 \implies \text{SafeHit}_{n,\Delta,a}(x) \implies \text{SameLane}_\Delta(x) = 1.} \quad (6.3)$$

Proof. The first implication is the identity-branch classification of Paper XXXI followed by (2.8). For the second, a carries each complete lane to a lane, while every guarded identity-branch point map preserves same-lane incidence. Hence every guarded a -return preserves that incidence. The target $a_{\#}^{-1}\widehat{\mathcal{T}}_{\Delta,\text{id}}$ is same-lane, so a state with marked tokens in different lanes cannot reach it.

It is tempting to conjecture

$$\text{Break}_\Delta(a) \neq \emptyset \implies \text{Hit}(a) = \text{Same}_\Delta. \quad (6.4)$$

Theorem 5.1 does not prove (6.4). A same-lane source may occupy a lane on which a is dihedral, while order breaking occurs only on another lane. Moving the complete five-token context between those lanes is a new guarded lane-mobility problem. An arbitrary placement cannot replace an actual path.

Inside the narrower affine normalizer family, the hostile descent question remains sharper than multiplier classification. One must test whether there exist

$$w, w' \in W_\Delta \quad \text{with} \quad \mu(w) = \mu(w') \quad \text{but} \quad \text{Prom}(w) \neq \text{Prom}(w'). \quad (6.5)$$

Such a pair would prove that phases or lane permutations are part of the smallest Safe-Hit observable within that subfamily. A failed search or an undefined branch would not prove non-descent.

Accordingly, the following remain open:

1. a lane-type-sensitive mobility theorem;
2. the full lane-stabilizer promotion dichotomy;
3. multiplier-only descent across affine phases and ordinary-lane permutations;
4. survivor-placement classification after the marked pair hits the target.

These questions are not needed for Theorem 5.1.

7 Computational Artifacts

The paper-owned evidence package is available under [experiments/paper32/](#) in the RIME repository. It has no dependency on the broader exploratory source tree.

Surface	Paper-owned path	Role
Bounded finite controls	<code>affine_five_token_audit.py</code> and <code>results/</code>	Replays the retained affine, arbitrary-permutation, and multi-lane hostile domains
Development validation	<code>development-manifest.json</code> and <code>validation/</code>	Binds and replays the eight-artifact paper-owned evidence closure
Release validation	<code>release-manifest.json</code> and the public receipt under <code>results/</code>	Binds the manuscript, reader PDF, bibliography, nested development closure, environment, and validators

The single-lane affine control covers $6 \leq n \leq 12$, every coprime kernel separation, every unit affine multiplier, every marked five-token state, every hole-deleted order fiber, and every applicable local gap transfer. This affine domain is a proper subfamily of the permutation theorem. A second control checks all 4,560 branch permutations in the one-lane domains $n = 6, 7$: 92 cycle automorphisms and 4,468 nonautomorphisms. A separate $6 \leq n \leq 9$ multi-lane scan enumerates all affine normalizer phases and ordinary-lane permutations in that bounded domain.

The finite audits are bounded consistency controls. They are not proofs of the all- n theorems and are not promoted as independent Computational Certificates. In particular, the multi-lane scan is a hostile search for multiplier non-descent; a bounded absence of variation is not an all- n descent theorem or a full-stabilizer result. The public receipt records performed replay and local closure verification, is excluded from its own closure, and is not an independent mathematical validation.

8 Claim Status and Boundary

All positive claims below have direct mathematical proofs in the manuscript. No finite census is a theorem premise.

Claim	Status	Scope
Exact skew quotient and target lifting	Imported	Paper XXXI normalizer interface
Pathwise identity simulation	Imported and restated as Lemma 2.1	Every branch permutation
Lane-wise dihedral classification	Proposition 3.1	Every lane-partition stabilizer branch whose lane restrictions are cycle-graph isomorphisms
Guarded unit gap transfer	Lemma 4.1	One-lane identity branch
Directed gap connectivity	Lemma 4.2	Every fixed weak-composition fiber
Order-fiber transitivity	Theorem 4.3	One-lane identity branch
Single-lane permutation dichotomy	Theorem 5.1	Every $a \in \text{Sym}(E)$
Lane-stabilizer Safe-Hit sandwich	Corollary 6.1	Every $a \in \text{Stab}(\mathcal{C}_\Delta)$
Bounded affine controls	Finite consistency control	$6 \leq n \leq 12$; proper affine subfamily only
Bounded arbitrary-permutation control	Finite consistency control	All 4,560 one-lane branches for $n = 6, 7$
Multi-lane hostile search	Finite consistency control	Affine normalizer branches for $6 \leq n \leq 9$
Full lane-stabilizer promotion classification	Open	Requires new mobility or obstruction
Survivor incidence	Not claimed	Marked-pair reachability is insufficient
Raw-to-typed transfer bridge	Not claimed	No new semantic fields or authority

The following implications are explicitly forbidden:

$$u \not\equiv \pm 1 \pmod{M} \not\Rightarrow \text{Prom}(w) \neq \emptyset, \quad (8.1)$$

because mixed-sign lane-wise dihedral multipliers need not create any promotion;

$$\text{Break}_\Delta(a) \neq \emptyset \not\Rightarrow \text{Hit}(a) = \text{Same}_\Delta \quad (8.2)$$

without a multi-lane mobility theorem; and

$$\text{raw Safe-Hit} \not\Rightarrow \text{typed transfer membership or projectability.} \quad (8.3)$$

9 Conclusion

The branch-permutation frontier has two distinct thresholds.

The first is lane-local, not group-global: a branch creates no new Safe-Hit state whenever it carries each arithmetic lane to a lane by a cycle-graph isomorphism. For affine normalizer branches this includes mixed Chinese-remainder signs excluded by the shorthand $u \equiv \pm 1 \pmod{M}$.

The second threshold is exact in the one-lane regime. There the guarded identity dynamics is transitive on every hole-deleted token-order fiber. A branch permutation outside $\text{Aut}(C_L)$ necessarily sends some physical nonedge to an edge. Mobility reaches a compatible nonadjacent five-token state, the globally enabled branch twist creates adjacency, and the identity simulation completes the target witness. Hence every state becomes target-reachable.

In compressed form,

$$\boxed{\text{guarded gap mobility} + \text{cycle-edge conversion} \implies \text{complete one-lane Safe-Hit.}} \quad (9.1)$$

The remaining multi-lane problem is now sharply separated. It is not another defect-classification problem. It asks whether a five-token spectator context can be moved, by actual guarded relations, to a lane on which the branch restriction breaks cyclic order. The affine multiplier question survives as one narrower descent problem inside the larger lane-stabilizer dynamics.

References

- [1] Mikhail V. Berlinkov and Cyril Nicaud. Synchronizing almost-group automata. *International Journal of Foundations of Computer Science*, 31(8):1091–1112, 2020.
- [2] David Fernando Casas Torres. Completely reachable almost group automata, 2024.
- [3] Ján Černý. Poznámka k homogénnym experimentom s konečnými automatmi. *Matematicko-fyzikálny časopis Slovenskej Akadémie Vied*, 14:208–216, 1964.
- [4] WuJun Chen. Branch-normalized partial returns in single-defect circular automata. RIME Paper XXXI, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23050809>.
- [5] WuJun Chen. Cyclic lineage dynamics and rank-five first-exit frontiers. RIME Paper XXIX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23028361>.
- [6] WuJun Chen. Universal return groups and punctured rotations. RIME Paper XXX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23034428>.
- [7] Henk Don. The Černý conjecture and 1-contracting automata. *The Electronic Journal of Combinatorics*, 23(3):P3.12, 2016.
- [8] Lucien Dubuc. Sur les automates circulaires et la conjecture de Černý. *RAIRO Informatique théorique et applications*, 32:21–34, 1998.
- [9] David Eppstein. Reset sequences for monotonic automata. *SIAM Journal on Computing*, 19(3):500–510, 1990.

- [10] Robert Ferens and Marek Szykuła. Recognizing completely reachable automata in quadratic time. *ACM Transactions on Algorithms*, 22(2):1–36, 2026.
- [11] François Gonze and Raphaël M. Jungers. On completely reachable automata and subset reachability. In *Developments in Language Theory*, volume 11088 of *Lecture Notes in Computer Science*, pages 330–341. Springer, 2018.
- [12] Andrew Ryzhikov and Anton Shemyakov. Subset synchronization in monotonic automata. *Fundamenta Informaticae*, 162(2–3):205–221, 2018.
- [13] Mikhail V. Volkov. Synchronizing automata and the Černý conjecture. In *Language and Automata Theory and Applications*, volume 5196 of *Lecture Notes in Computer Science*, pages 11–27. Springer, 2008.
- [14] Mikhail V. Volkov. Synchronization of finite automata. *Russian Mathematical Surveys*, 77(5):819–891, 2022.
- [15] Petra Wolf. Synchronization under dynamic constraints. In *40th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2020)*, volume 182 of *Leibniz International Proceedings in Informatics*, pages 58:1–58:14. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2020.